



General method of wavelength-tuning of Dirac-like cones in photonic crystals

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Abstract: Near-zero refractive index (NZI) materials offer a wide range of applications, from classical optics to quantum technologies. The NZI regime can be effectively achieved by engineering the dispersion of photonic crystals to form Dirac-like cones at the Brillouin zone center. This work presents a machine-learning-driven framework for systematically identifying and tuning Dirac-like cones across arbitrary material systems and operating wavelengths. By employing a batched Bayesian optimization strategy, we efficiently map multi-parameter design spaces to locate specific geometric configurations that achieve accidental triple degeneracy. Our findings demonstrate that the dielectric filling factor governs the transition from quadratic to linear Dirac-like dispersion by modulating the inter-modal coupling strength. Expanding to three-dimensional simulations, we establish design rules to tune this degeneracy to specific target wavelengths such as 1550 nm. Ultimately, this versatile approach provides a scalable pathway for developing designs of low-loss, near-zero-index materials for advanced optical applications.

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1. Introduction

Materials with a near-zero refractive index (NZI) exhibit extraordinary phase-change-free propagation of electromagnetic waves and have attracted strong interest in recent years [1]. A wide range of applications is possible; for example, they enhance nonlinear processes such as third-harmonic generation [2,3], they can be used as coherent perfect absorbers [4], and can be exploited for electromagnetic cloaking of objects of arbitrary shapes [5,6] or as super-reflectors [6]. Devices such as lasers [7] and antennas [8] benefit from the NZI-based implementations by utilizing the uniform spatial phase to achieve large-area, single-mode emission and highly directive radiation, respectively.

More recently, their applications in quantum optics have stimulated significant research. The NZI super-coupling effect [9,10] enables the coupling of quantum emitters over long distances and the realization of on-chip optical quantum networks [11]. NZI cavities can be exploited to tailor the far-field cooperative dynamics between embedded quantum emitters and achieve Dicke superradiance [12,13] or as multi-port hubs for non-reciprocal quantum devices [14].

The near-zero-index operation regime can be accessed by tuning the operating frequency close to the plasma frequency of metals [1]. With metals, this approach however is affected by high losses, and, for this reason, an all-dielectric platform with low loss would be favorable. In fact,

photonic crystals [15] can be used to obtain an effective NZI condition if a Dirac-like cone dispersion is designed in the center of their Brillouin zone [16]. This dispersion is achieved by the triple degeneracy of three bands in the Γ -point ($k_x, k_y = 0$): two linearly-dispersive bands and a flat one. It has however been proven that not all photonic crystals with a Dirac-like cone can be mapped to homogeneous NZI materials, because a set of additional conditions needs to be satisfied simultaneously: (i) the degeneracy point is formed by monopole and dipole moments, (ii) the degeneracy is at low enough frequency, and (iii) there is only one propagating band in the extended states above the triple degeneracy frequency ω_D [17].

Regardless of the target application, the essential first step, which is the primary focus of this article, is the identification of a photonic crystal structure that exhibits the requisite triple degeneracy to form a Dirac-like cone tailored to a specific material platform and operating wavelength. This is done by tuning the photonic crystal parameters, such as the hole or pillar radii, within the periodically repeated unit cell. In the simplest cases, where only one parameter is changed while the others are fixed, it is sufficient to sweep its value and select the one corresponding to triple degeneracy [5,18]. When two parameters are used, a two-dimensional sweep can still be used to map the degeneracy [19]; however, this method is computationally expensive because it requires calculating the photonic crystal eigenmodes for a large number of parameter combinations in high-resolution maps.

More recently, inverse design has been applied to the problem of finding degeneracy points in a multi-parameter space [20]. However, the specific parameter combinations providing the target dispersion have not been systematically mapped.

In this work, we consider a parametrized photonic crystal structure – whose geometric features are defined by a set of independent variable parameters rather than static dimensions – that allows accidental degeneracy of modes with symmetries that satisfy the selection rule for the formation of Dirac-like cones [21,22]. Unlike degeneracies forced by lattice symmetry, these accidental points occur when independent modes are specifically tuned to coincide in frequency [5]. We use inverse-design machine-learning-powered methods to efficiently generate a surrogate map of all the degeneracy points obtained by combining the different photonic crystal parameters. While our initial mapping reveals a continuous locus of triple-degeneracy points, we find that not all points support true Dirac-like cones, even if they satisfy the necessary symmetry selection rules. To overcome this, we perform a post-mapping analysis to isolate the specific configurations that exhibit true linear dispersion. We validate the framework's versatility across different parametrized configurations. Finally, by scaling the 3D photonic crystal structure, we demonstrate the possibility of systematically tuning these isolated cones to specific wavelengths.

2. Methods

2.1. Selection rules for the formation of the Dirac-Like cone in photonic crystals

It has been shown that Dirac-like cones are closely linked to mode symmetries [5,21,23]. Hence, the first step toward realizing such a platform is to identify a photonic crystal structure that supports Dirac-like cones. A structure can be considered as the combination of a lattice and a basis [15,24]: the former is the grid of points that defines the periodicity of the crystal, while the latter is the geometric pattern - the dielectric distribution $\epsilon(\mathbf{r})$ - that is repeated at each lattice point and is contained within a unit cell. What matters is the point group of the crystal structure: this is given by the intersection of the symmetries belonging to the point groups of the lattice and the basis [25,26]. For each point group, only the accidental degeneracy of certain irreducible representations (irreps) [26] at the Γ -point can generate a Dirac-like cone [23,27]. For the C_{4v} point group, characteristic of a square lattice, and for the C_{6v} point group, characteristic of a triangular lattice, these combinations are reported in table 1: the shapes of the dispersion curves are indicated, where D, DD, and Q denote the Dirac cone, double Dirac cones, and the quadratic dispersion surface, respectively. The Dirac-like cone is labeled by D + Q.

Table 1. Types of dispersion curves generated by the accidental degeneracy of the photonic crystal modes

C_{4v} symmetry		
Mode 1	Mode 2	Dispersion
E	A_1, A_2, B_1, B_2	D + Q
A_1, A_2, B_1, B_2	A_1, A_2, B_1, B_2	2Q
E	E	4Q
C_{6v} symmetry		
Mode 1	Mode 2	Dispersion
E_1	E_2	DD
E_1	E_1	4Q
E_2	E_2	4Q
E_1	A_1, A_2	D + Q
E_2	B_1, B_2	D + Q
E_2	A_1, A_2	3Q
E_1	B_1, B_2	3Q
A_1, A_2, B_1, B_2	A_1, A_2, B_1, B_2	2Q

The band structure of the bulk photonic crystal can be computed using *MIT Photonic Bands* [28]. As soon as the bands with the right symmetries are found, we can optimize the parameters of the photonic crystal, such as the size of the scatterer, and determine whether a locus of points exhibiting triple degeneracy exists. Diving deeper into the selection rules for the formation of Dirac-like cones, we see that the degenerate orthonormal Bloch modes $\{\varphi_i(\mathbf{r})\}$ need to have a non-vanishing mode coupling-integral [21]:

$$p_{ij} = \frac{i}{A} \int_{\text{unit cell}} \frac{1}{\varepsilon(\mathbf{r})} \left[\varphi_i^*(\mathbf{r}) \nabla \varphi_j(\mathbf{r}) - \varphi_j(\mathbf{r}) \nabla \varphi_i^*(\mathbf{r}) \right] d\mathbf{r}, \quad (1)$$

where A is the unit cell area (or volume in a three-dimensional study). Some irreducible representations, given their symmetries, cannot satisfy this necessary condition, and they can be excluded from the set of candidates for the generation of the linear dispersion even if they are degenerate at Γ [21].

2.2. Batched Bayesian optimization

To find these points, we define the cost function,

$$C(\mathbf{x}) = \frac{\omega_{\text{top}}^{\Gamma}(\mathbf{x}) - \omega_{\text{bot}}^{\Gamma}(\mathbf{x})}{\omega_{\text{mid}}^{\Gamma}(\mathbf{x})}, \quad (2)$$

which is evaluated by numerically isolating the normalized frequencies of the target bands at the Γ point - $\omega_{\text{bot}}^{\Gamma}$, $\omega_{\text{mid}}^{\Gamma}$, $\omega_{\text{top}}^{\Gamma}$, from smaller to bigger value - for a parametrized crystal structure with parameters $\mathbf{x} = (x_1, \dots, x_n) \in \mathcal{S}_{\mathbf{x}}$. Low values of the cost function correspond to triple degeneracy. The target bands (top, mid and bot) can be defined directly by providing the index that orders them from lowest to highest frequency at Γ . Alternatively, they can be identified automatically and dynamically using the symmetry eigenvalue functionality in *MPB* to determine the irreducible representations (irreps) [26] of each band [29], and matching them against the

desired set that produces D+Q dispersion as in table 1. The latter is a safer approach against band swapping.

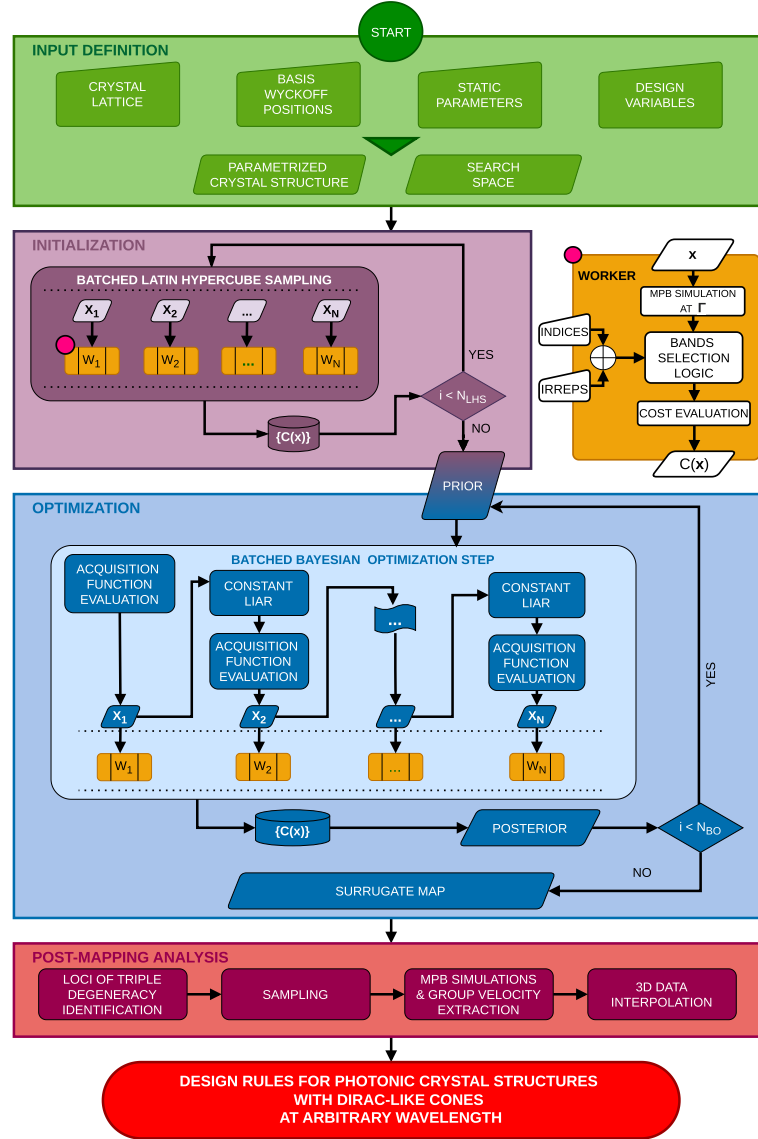


Fig. 1. Workflow of the machine-learning-driven method for discovering and tuning Dirac-like cones. The process requires the crystal lattice and the Wyckoff positions of the scatterer as inputs, along with the static parameters and design variables. These inputs define the parametrized crystal structure. The batched Latin Hypercube Sampling builds the initial surrogate model (prior), followed by a batched Bayesian Optimization loop that evaluates a bandgap cost function $C(x)$ using *MIT Photonic Bands (MPB)* solver. During this stage, target bands are identified such that they satisfy the symmetry selection rules required for the formation of Dirac-like cones; we can do this by matching irreducible representations (irreps) via MPB. The surrogate map obtained from the posterior at the end of the loop identifies loci of triple degeneracy, which are then analyzed for group velocity to isolate Dirac-like cones and ultimately establish design rules for arbitrary target frequencies.

In general, a closed-form solution of eq. (2) is not known, and it can be assumed to be a black box. Given that the computation of the cost function can also be expensive to evaluate, especially for high-resolution three-dimensional simulations, we employed a batched Bayesian Optimization (BO) strategy [30–32]. The standard BO implementation is a sequential process that begins with a prior distribution representing the initial beliefs about the cost function. As evaluations are completed, this is updated to a posterior distribution (the surrogate model). The user must define the acquisition function (AF), which leverages the posterior mean and variance to balance exploration of the parameter space and exploitation of the current model.

We want to emphasize that our focus is not on the global minimum of eq. (2), but rather on characterizing the region of gap closure within the parameter space S_x . The parallel batched implementation of BO is well-suited to this purpose. In our implementation, based on [33], we leverage the DTU HPC [34] cluster to concurrently evaluate the cost function for a batch of points, using multiple workers. With the first generations of points, we uniformly map the space using the Latin Hypercube Sampling (LHS) method [35], and the optimizer automatically builds the initial surrogate model [33]. The points in the following generation are determined using the lower confidence bounds (LCB) AF with constant liar (CL). This means that the first point of the batch, $\tilde{x}$, is found calculating the LCB acquisition function, then a temporary objective value $C^{CL}(\tilde{x}) = \min\{C(x)\}$ is assigned to it, so that another point in the batch can be determined using the temporary updated surrogate model [36]. When the stop condition is met, the surrogate map shows the locus of the point of triple degeneracy, if any.

For the subsequent analysis and visualization of the surrogate maps, we define our figure of merit as the inverse of the expectation $\mathbb{E}[C(x)]^{-1}$, which asymptotically diverges at the exact points of triple degeneracy, allowing for precise identification of the gap-closing regions.

The computational workflow is depicted in fig. 1. Following the generation of the surrogate map, samples demonstrating triple degeneracy are selected for additional band dispersion analysis, as discussed further in section 3.

3. Results

3.1. Indium phosphide photonic crystal slab with holes

3.1.1. 2D simulations

The first photonic crystal structure we have considered is realized from an Indium Phosphide (InP) slab, with a square lattice and a diatomic base of air holes of different sizes occupying the Wyckoff positions $1a$ and $2b$ of the point group C_{4v} [26,37]. For the 2D case, there are only two parameters, r_1 and r_2 , respectively, the radii of the holes in $1a$ and $2b$. The band diagram of this structure, for an arbitrary choice of its parameters, is reported in fig. 2.

We focus on this structure because it supports embedded InAs/InP quantum dots emitting in the 1550 nm Telecom C band [38]. An effective near-zero-index cavity based on a system of diamond pillars has been theoretically shown to amplify the collective quantum dynamics of Nitrogen-Vacancy centers [13]; therefore, translating these effects to the Telecom C band demands crystal configurations tailored for the InP platform.

Given that epitaxial growth quantum dots are disk-shaped, they will have a strong in-plane dipole moment, enabling them to couple effectively with TE modes. For this reason, we focus on the triple-degeneracy of the TE modes. Observing fig. 2, we can see the presence of the degenerated E modes at the center of the Brillouin zone (modes 4 and 5, starting counting from mode 1 at the bottom of the band diagram): by tuning the parameters, the accidental degeneracy might happen between them and the ones that are close enough (modes 3 and 6). At the center of the Brillouin zone, all these modes have the right irreducible representations to satisfy the selection rules for generating a Dirac-like cone ($D + Q$) reported in table 1.

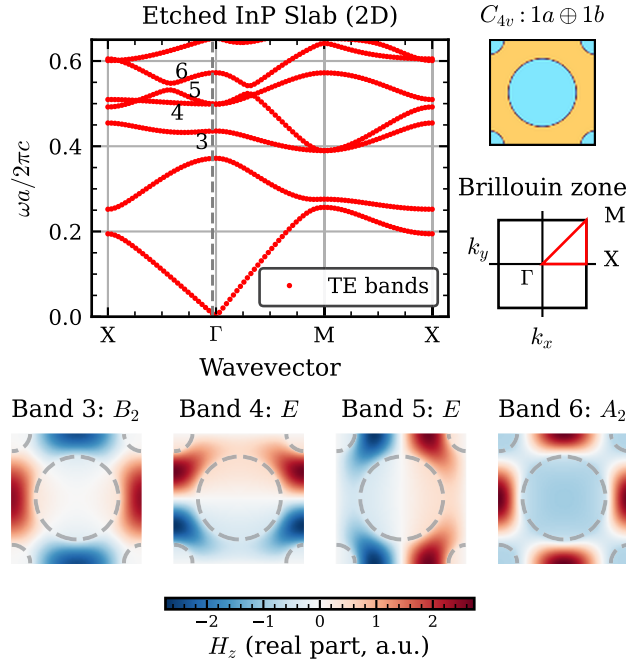


Fig. 2. Simulation results of the 2D unoptimized photonic crystal slab. **Top-left:** Photonic band diagram for TE (red) modes; the vertical dashed line indicates the wavevector selected for field visualization, $\Gamma + \delta k$. **Top-right:** Real-space dielectric distribution for the C_{4v} lattice with holes in the Wyckoff positions $1a \oplus 1b$; below, the corresponding Brillouin zone with the irreducible path highlighted. **Bottom:** Transverse electric (TE) eigen-field profiles (H_z component) for bands 3–6 at the selected k -point where the degeneracy is broken and bands can be enumerated unambiguously. The mode labels denote the irreducible representations at Γ for each branch. Note that the E representation is twofold degenerate at Γ .

Using the batched Bayesian Optimization strategy described in section 2.2, as shown in fig. 3, we managed to find a bi-dimensional locus of points in the parameter space by targeting modes 4, 5 and 6. An ellipse has been fitted to the data to highlight the degeneracy ridge [39].

Two of these points, highlighted in fig. 3, have been further analyzed, and the results are reported in fig. 4(a-d). These points were chosen by evaluating the filling factor (FF), defined as the fraction of the 2D unit cell occupied by the dielectric material:

$$\text{FF} \equiv \frac{A_{\text{dielectric}}}{A_{\text{unit cell}}} = 1 - \pi \frac{r_1^2 + r_2^2}{a^2}. \quad (3)$$

By defining an arbitrary threshold for this value, we selected the data point associated with the lowest cost. We observed that even in the presence of the same irreps (cf. figure 4(c) and (d)), the dispersion around the Γ point is a Dirac-like cone only for the photonic crystal with a smaller filling factor (compare fig. 4(a) and (b)). For completeness, we compared the band diagram obtained from simulations at different k -points with the results of the $\mathbf{k} \cdot \mathbf{p}$ perturbation theory [21], computed from the eigenmodes at Γ . When 9 eigenmodes are used for the computation of the $\mathbf{k} \cdot \mathbf{p}$ expansion, the dispersion closely matches *MPB* results. In contrast, when only the degenerate modes are used for the computation, the nonlinearities of the dispersion can not be correctly predicted.

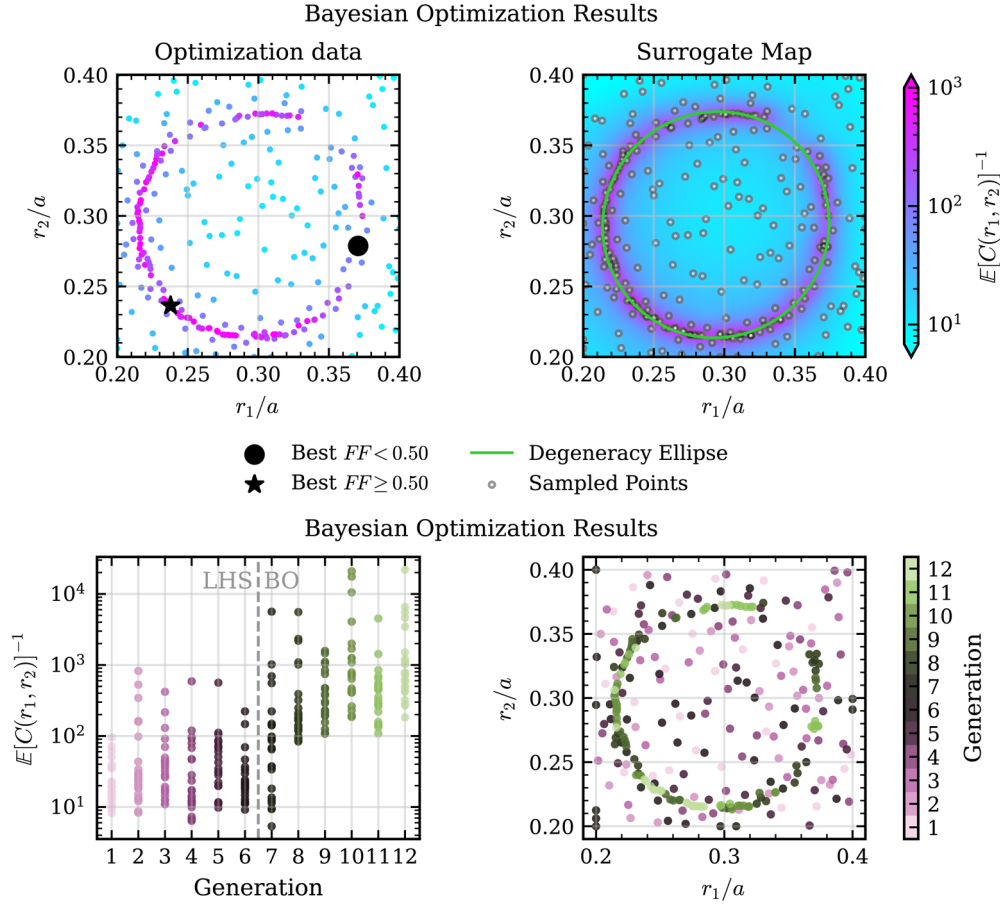


Fig. 3. Bayesian optimization analysis for the 2D photonic crystal slab. **Top:** Final optimization results, displaying the raw data points colored by the inverse of the cost expectation and the corresponding surrogate model predictions (right). The green ellipse indicates the fitted degeneracy ridge. **Bottom:** Optimization process history, showing the cost function convergence on a logarithmic scale (left) and the parameter space sampling distribution (right). The transition from LHS to the BO strategy balances exploration of the space and exploitation of the model, enabling the acquisition function to successfully identify and focus on the degeneracy locus.

To gain a deeper understanding of these results, we selected sample points along half of the degeneracy ellipse (due to the symmetry of this parametrized photonic crystal) and fine-tuned r_2 to ensure the modes remained strictly (fig. 4(a)). Certain points were excluded where the dielectric structure became so narrow that the solver treated them as disconnected regions. For each point, we computed the target bands at a distance of $||k||a/2\pi = 0.1$ from the Γ -point, along the $\Gamma \rightarrow X$ and $\Gamma \rightarrow M$ directions.

For our analysis, we evaluated the group velocity of band 6 at Γ using two methods: directly with MPB via the Hellmann-Feynman theorem [15], and using perturbation theory [21]. In the latter case, the velocity was computed only in the $\Gamma \rightarrow X$ direction, considering only the degenerate modes.

We observed a good match between the two approaches, as illustrated in fig. 4(a), where the group velocity is plotted against the dielectric filling factor of the crystal structure.

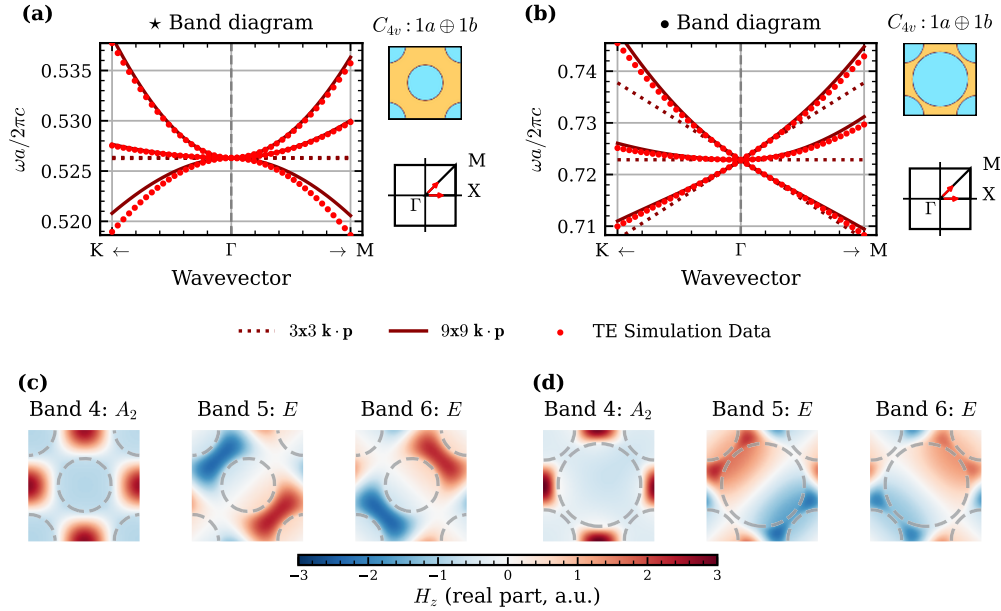


Fig. 4. Accidental degeneracy in the 2D photonic crystal slab for two different dielectric filling factors. **(a,c - ★)**: Case with a high dielectric filling factor. In this regime, all dispersions near the Γ -point are quadratic, and the Dirac-like cone is not formed. **(b,d - ●)**: Case with a small dielectric filling factor. In the vicinity of the Γ -point, the band dispersion exhibits a Dirac-like cone, characterized by two bands with linear dispersion and one flat band (with quadratic dispersion only), $D + Q$. **(a,b)**: The band dispersion obtained from *MPB* (red dots) is compared with that predicted by perturbation theory (solid and dotted lines). **(c,d)**: Degenerate eigenmodes at Γ ; for both simulations $k_{max}a/2\pi = 0.1$. The TE eigen-field profiles are displayed for each case.

Consistent with our previous findings, we observed that the group velocity can be increased by tuning the filling factor of the parametrized photonic crystal (fig. 5(a,b)). However, the group index $n_g = c/v_g$ remains consistently greater than the refractive index of the bulk InP slab (fig. 5(b)). The reduction in group velocity is linked to an increase in the Bloch wave impedance, which diverges as $v_g \rightarrow 0$ [40]. A substantial impedance mismatch complicates the efficient coupling of light from external sources, such as standard defect waveguides, into near-zero-index photonic crystal cavities unless specialized transition regions are introduced.

The introduction of the filling factor is particularly advantageous for this analysis, as it allows us to reduce the dimensionality of the problem from $\{r_1, r_2\}$ to the single variable FF.

As shown in fig. 5(d), the group velocity is directly proportional to the inter-modal coupling strength, calculated using eq. (1) as:

$$\text{Coupling Strength} \equiv \frac{\sqrt{\sum_{i \neq j} |p_{ij}|^2}}{\omega_D}, \quad (4)$$

which is the quantity that inherently vanishes when the symmetry selection rules are not satisfied (fig. 5(d)).

For each direction, we performed a third-order polynomial fit on the higher-frequency band:

$$\omega_{\text{FIT}}(k) = a_0 + a_1 k + a_2 k^2 + a_3 k^3, \quad (5)$$

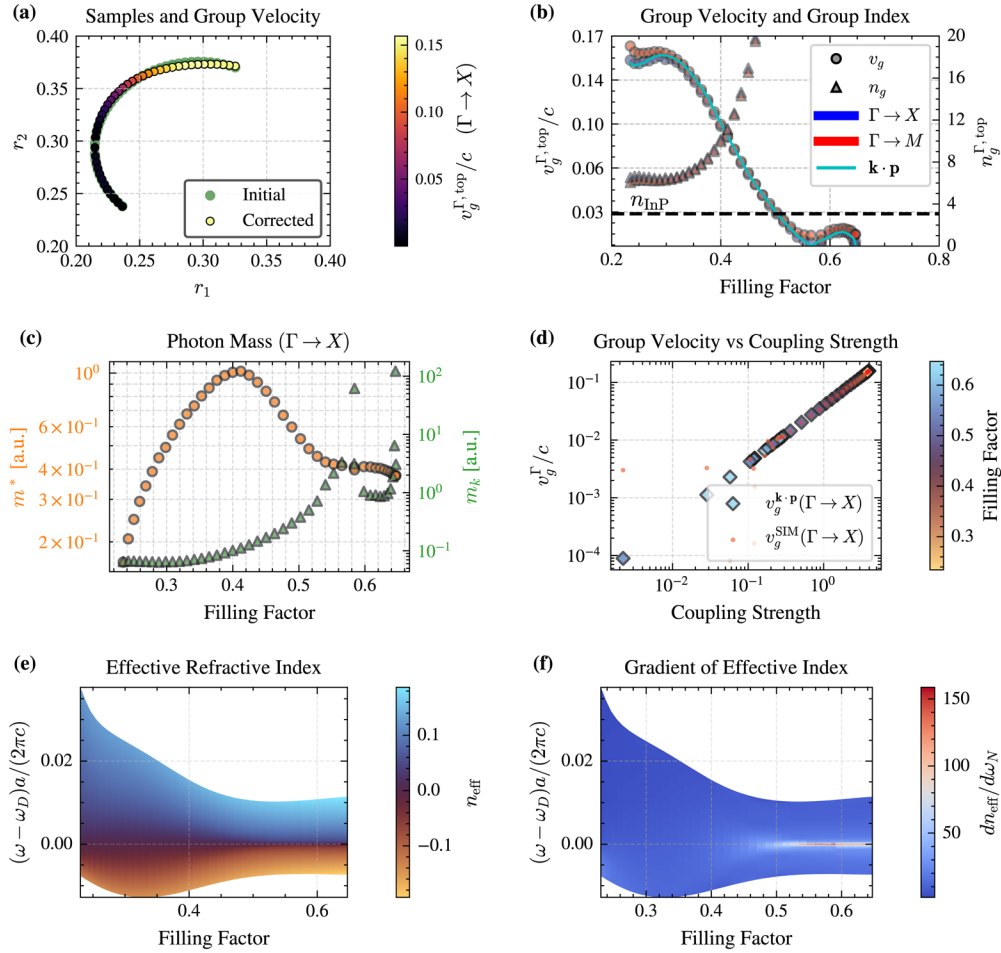


Fig. 5. Evolution of modal properties and dispersion metrics near the Γ point. **(a)** Sample Points in $\{r_1, r_2\}$ -space, before (green) and after degeneracy correction. The colormap of the corrected points shows the group velocity ($v_g = ||\mathbf{v}_g||$) of the top band. **(b)** Group velocity and group index (n_g) against the filling factor, evaluated along the $\Gamma \rightarrow X$ and $\Gamma \rightarrow M$ directions from the top band. The dashed line represents the bulk refractive index of the substrate, n_{InP} . **(c)** Linearity metrics of the upper band dispersion illustrating the transition from heavy photons to massless photons by decreasing the filling factor, quantified via the curvature mass (m^*) and the kinetic mass (m_k). **(d)** Dependence of the group velocity on the coupling strength, comparing direct numerical simulations (v_g^{SIM}) with predictions derived from perturbation theory ($v_g^{k \cdot p}$). **(e)-(f)** Effective refractive index and its gradient with respect to the normalized frequency $\omega_N = \omega a / 2\pi$ along the $\Gamma \rightarrow X$ direction, calculated considering the top and the bottom bands. Data are obtained from a set of simulations exhibiting accidental degeneracy, swept across the filling factor with a fixed boundary of $k_{\text{max}} a / 2\pi = 0.1$.

where $k = ||\mathbf{k}||$. We used this fit to evaluate the curvature of the band at the Γ -point, yielding our first linearity metric, the curvature photon mass:

$$m^* \equiv \hbar \left(\frac{\partial^2 \omega}{\partial k^2} \right)^{-1} \equiv \frac{\hbar}{2a_2}. \quad (6)$$

To complement this, we introduced a second metric: the photon kinematic mass, defined as the ratio of momentum to group velocity:

$$m_k \equiv \frac{\hbar k}{v_g} = \frac{\hbar k}{\frac{\partial \omega}{\partial k}}. \quad (7)$$

While the curvature mass m^* characterizes the shape of the band (assuming a high value when the band is very linear), the kinematic mass m_k measures the transport inertia of the wavepacket (converging to exactly zero at the Γ -point for massless states).

The evolution of these two metrics is reported in fig. 5(c). For intermediate filling factors (FF ≈ 0.4), m^* reaches a distinct peak while m_k remains relatively small, signifying the region where the band exhibits the straightest, most Dirac-like dispersion. Conversely, at low filling factors (FF < 0.3), the drop in m^* implies a return to more curved bands. At higher filling factors (FF > 0.55), the kinematic mass spikes dramatically; because the modes cannot couple well through their gradients (eq. (1)), the group velocity collapses, pushing the system into a slow-light regime characterized by heavy effective photon masses and high chromatic dispersion.

The effective index measures the phase advance within the structure and is defined as:

$$n_{\text{eff}} = \frac{ck}{\omega}. \quad (8)$$

The signature of the Dirac-like cone is visible in the effective index along the $\Gamma \rightarrow X$ direction (fig. 5(e)): the index approaches zero near the Dirac frequency (ω_D). As expected, n_{eff} changes abruptly with respect to frequency in the high-filling-factor regime, where the massive gradient observed in fig. 5(f) indicates huge chromatic dispersion.

3.1.2. 3D simulations

So far, we haven't considered the actual value of the degeneracy frequency, because thanks to the scaling rules of photonic crystals [15] we can scale the lattice constant a to fix the degeneracy frequency to a desired value. However, when a slab of finite thickness is considered, the dimension of the parameter space increases, potentially altering both the shape and the size of the locus of points. As we explain below, a designer needs to carefully choose the parameters to ensure that the degeneracy occurs at the target frequency.

We started again from the geometry proposed in fig. 2(f), characterized by the parameters r_1 and r_2 . This time, we defined a supercell with a vertical periodicity of $4 \times a$ and a finite slab thickness h . For this scope, we targeted the TE-like bands 8, 9, and 10. For simplicity, we performed Bayesian optimization multiple times, sweeping the slab-to-lattice-constant ratio, h/a . In principle, our method can be readily extended to a higher-dimensional parameter space, but more points need to be computed to realize an accurate surrogate map of the degeneracy. From the surrogate model, we obtained, as before, an ellipse approximating the locus of degeneracy in each h/a -cut plane, as reported in fig. 6(a).

By fitting the top band in the vicinity of the Γ -point with a third-order polynomial, we computed the group velocity demonstrating again that the linearity of the Dirac-like cone depends on the filling factor (fig. 6(c)).

We can use physical units to clearly see that a thicker slab is needed to achieve a Dirac-like cone at longer wavelengths when the thickness-to-lattice-constant ratio is fixed. The degeneracy coverage, in fig. 6(d), shows the portion of the spectrum where it is possible to achieve degeneracy; for each of these bands, the short-wavelength side has to be chosen to get a high group velocity.

Finally, by fitting these data, we derived a group of design rules, shown in fig. 7(a), for the realization of Dirac-like cones in InP photonic crystal slabs at 1550 nm. The designer can arbitrarily choose the slab thickness; by tracking a vertical line, the values of a can be directly

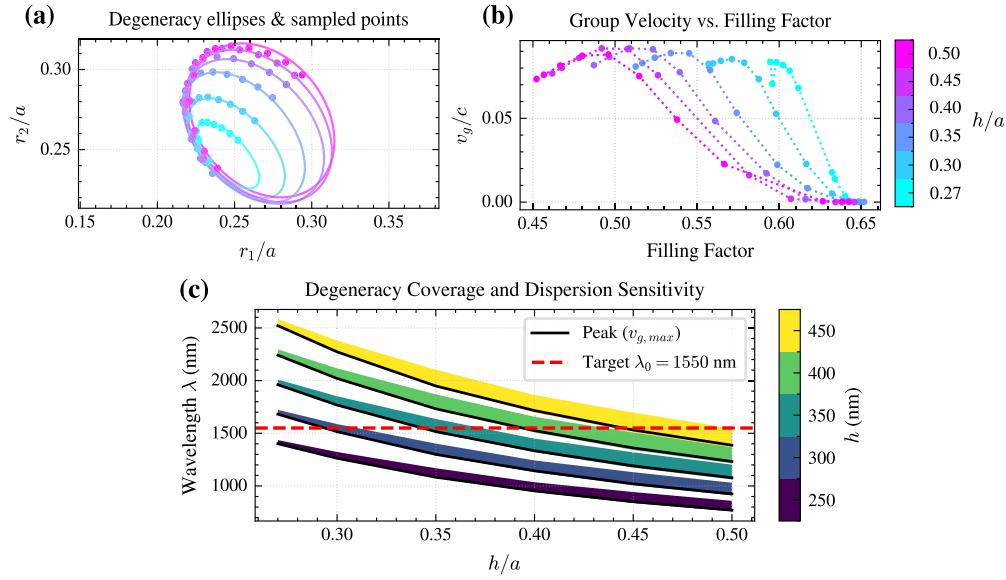


Fig. 6. Characteristics and degeneracy coverage of a finite-thickness slab structure ($\varepsilon = 9.46$). **(a)** Degeneracy ellipses in the $\{r_1, r_2\}$ -parameter space for various normalized slab thicknesses (h/a). Scattered points represent samples fine-tuned to maintain the strict degeneracy necessary for the following analysis. **(b)** Scaling of the group velocity (v_g) with the filling factor. **(c)** Degeneracy coverage and dispersion sensitivity as a function of normalized thickness. High group-velocity Dirac-like cones (solid black lines) are located on the short-wavelength side of these tuning bands. The horizontal dashed line marks the target telecom wavelength of $\lambda_0 = 1550$ nm.

obtained. The values of r_1 and r_2 are obtained from the coordinates of the corresponding point in polar coordinates:

$$\rho = \sqrt{r_1^2 + r_2^2}, \quad \theta = \frac{180^\circ}{\pi} \operatorname{atan}(r_2/r_1). \quad (9)$$

We observe that a thinner slab is required to increase the in-plane filling factor; this trade-off must be carefully considered to ensure the mechanical stability of the membrane during fabrication. We validated our design rules by computing the band diagram using the set of parameters corresponding to $h = 375$ nm, confirming, in fig. 7(b), the presence of a Dirac-like cone.

3.2. Photonic crystal made of silicon pillars

A system of silicon pillars has been considered to show that our method can also work for a very different system, with a hexagonal lattice and TM modes. We consider pillars of two different sizes, r_1 in the Wyckoff position 1a (center of the hexagonal cell in fig. 8(a)), and r_2 in the Wyckoff position 3c of the point group C_{6v} (middle point of the edges) [26,37]. In this case, we look for the accidental degeneracy between the low-frequency monopole and dipoles, corresponding to the irreps A_1 and E_1 [26]; these modes are often mapped to an NZI material [17,21]. The results of the BO optimization, with automatic irreps tracking, are shown in fig. 8(b) where two degeneracy branches are identified in the surrogate map.

Once again, we found that the group velocity depends on the filling factor, as reported in fig. 8(c), but, for this configuration, all of the points in both branches present the high group velocity that characterizes the Dirac-like cones.

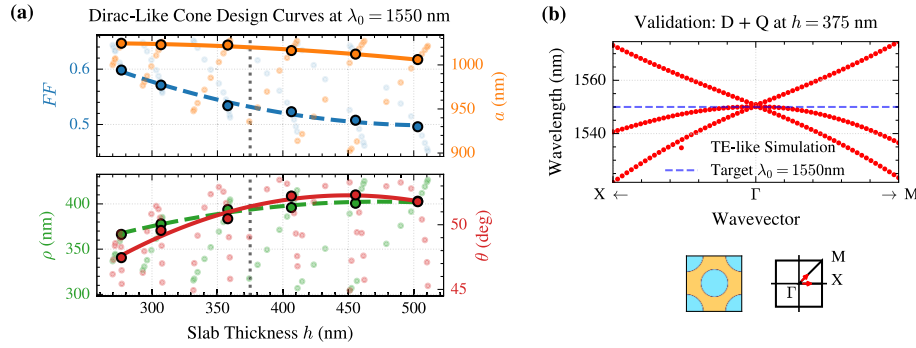


Fig. 7. Design rules and validation for an InP Photonic Crystal slab exhibiting Dirac-like cone dispersion. **(a)** Design rules for the realization of the target dispersion, $D + Q$, at $\lambda_0 = 1550$ nm. The curves illustrate the required lattice constant a , in-plane filling factor (FF), and size of the holes (ρ, θ) as a function of the slab thickness h . The highlighted points correspond to the photonic crystal samples with the highest group velocity at $\lambda_0 = 1550$ nm, and they have been interpolated to obtain the design curves. **(b)** Band diagram for an optimized 375 nm thick InP slab ($a = 1020$ nm, $r_1 = 246$ nm, $r_2 = 308$ nm), corresponding to the vertical line in (a). The simulation confirms the existence of a Dirac-like cone at the target wavelength of 1550 nm at the Γ point.

It is worth mentioning that in the second branch (fig. 8(b,c)), these modes are among the ones with the lower frequency (bands 2, 3, and 4) and this is another important condition for mapping their Dirac-like dispersion to effective NZI materials [17]. In the first branch, they are the 5th, 6th, and 7th bands.

The band diagram for one of these points is shown in fig. 8(d), and the corresponding eigenmodes are shown in fig. 8(e). The triple degeneracy occurs between a monopole and two dipoles at Γ , and inspection of the mode profile far from the center of the Brillouin zone shows that the strong coupling between the modes makes bands 2 and 4 very similar.

4. Conclusions

This research has established a robust, machine-learning-driven framework for systematically identifying and tuning the frequency of Dirac-like cone dispersions in all-dielectric photonic crystals. By employing a batched Bayesian Optimization strategy, we efficiently mapped the multi-parameter space of photonic crystal structures with different characteristics to identify the specific geometric configurations required to achieve accidental triple degeneracy.

Our results demonstrate that while point group symmetries provide the fundamental selection rules for degeneracy, the transition between linear Dirac-like dispersion and purely quadratic surfaces is critically governed by the dielectric filling factor. This factor modulates the inter-modal coupling strength, though its impact varies across configurations. In the square lattice InP slab, high filling factors lead to a reduced coupling and quadratic dispersion. Conversely, the hexagonal silicon pillar lattice exhibits relatively high group velocity and robust linear dispersion across all samples. Consequently, the filling factor serves as a critical design knob for maintaining band linearity, especially in configurations where mode coupling depends strongly on the dielectric distribution.

To bridge the gap between theoretical models and physical devices, we expanded our analysis to three-dimensional simulations to capture the impact of finite slab thickness. By analyzing samples from the optimization step, we derived actionable design rules to tune the degeneracy wavelength of Dirac-like cones. These rules consist of predictive curves mapping a chosen slab thickness (h) to the required lattice constant (a) and hole dimensions (r_1, r_2 , represented via polar

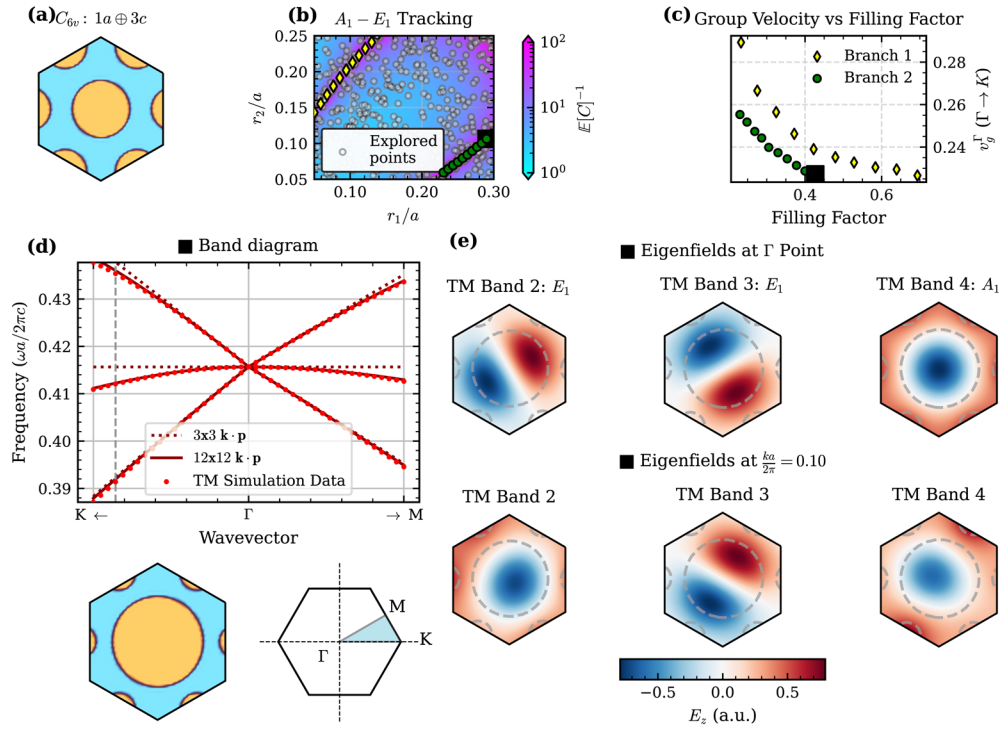


Fig. 8. Accidental degeneracy and Dirac-like cone formation in a hexagonal lattice of silicon pillars for transverse magnetic (TM) modes. (a) Hexagonal unit cell detailing pillar placement at Wyckoff positions 1a (center, with size r_1) and 3c (edge, with size r_2). (b) Surrogate map generated via Bayesian optimization tracking the $A_1 - E_1$ degeneracy, revealing two distinct solution branches. Branch 1 involves the accidental degeneracy of bands 5, 6, and 7, while Branch 2 targets the lower-frequency bands 2, 3, and 4, counting from the lowest frequency. (c) Group velocity as a function of the dielectric filling factor for both degeneracy branches. (d) Band diagram for an optimized configuration ($r_1 = 0.290a$, $r_2 = 0.106a$) demonstrating triple degeneracy at the Γ -point. Results of the perturbation theory are compared with the dispersion obtained with *MPB*. (e) TM eigenfield distributions at and slightly off from the Γ -point, confirming the coupling between monopole (A_1) and dipole (E_1) modes.

coordinates ρ and θ) to achieve a triple degeneracy yielding a highly linear dispersion at the desired wavelength.

Importantly, while these specific design curves were generated for a particular parametrized InP photonic crystal slab targeting a 1550 nm wavelength and a particular set of irreducible representations, the underlying approach is universal because it does not rely on any system-specific assumptions. This framework can be applied to diverse material platforms and photonic crystal architectures at any arbitrary target wavelength. This approach was supported by successfully realizing a Dirac-like cone in a 375 nm-thick slab simulation, in a configuration not previously analyzed, confirming that the methodology effectively scales to meet specific experimental requirements.

Future work is needed to experimentally validate these findings and to verify the performance of these designs in physical implementations. Additionally, the cost function could be modified to include the group velocity, enabling a unified analysis that simultaneously optimizes for both triple degeneracy and band linearity.

Considering that in practice it can be advantageous to create a Dirac-like cone in a frequency window with no losses and, thus, flat material dispersion, our lossless approach is reasonable, and our simulations don't account for material dispersion. When included, the material dispersion does not affect the group velocity at the gamma point, which is determined by the coupling of the degenerate modes; however, it introduces additional shifts in the photonic crystal bands far from Γ point, which can be estimated using a perturbative approach. On the other hand, adding losses alters that Dirac-like cone into a flat band [41]. To accurately simulate these effects, a different solver should be used, provided that the bands can be indexed and their movement tracked in the frequency-parameter space.

Although we used simple circles for our photonic crystals, our method can also optimize complex shapes with many degrees of freedom. However, these shapes must be carefully designed to ensure they do not reduce the photonic crystal's symmetries to a subgroup of the target symmetries, because the point group determines the irreducible representations of the modes at the Γ point and the selection rules for the formation of a Dirac-like cone.

Ultimately, the surrogate maps generated through this process serve as a versatile design tool for tailoring band dispersion, providing a scalable pathway toward the development of low-loss NZI materials for applications ranging from enhanced nonlinear signal generation to on-chip optical quantum networks.

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Data availability. Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

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